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▶ **To cite this version:**

Mathilde Husky, Christophe Leon, Enguerrand Du Roscoat, Helen-Maria Vasiliadis. Prevalence of past-year major depressive episode among young adults between 2005 and 2021: Results from four national representative surveys in France. Journal of Affective Disorders Reports, 2023, 342, pp.192-200. ⟨10.1016/j.jad.2023.09.019⟩. ⟨hal-04583212⟩

HAL Id: hal-04583212

https://hal.science/hal-04583212v1

Submitted on 1 Oct 2025

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Version of Record: https://www.sciencedirect.com/science/article/pii/S0165032723011527
Manuscript_6e7bd7c97279a032486b7dd6fbf578d9

Prevalence of past-year major depressive episode among young adults between 2005 and 2021: Results from four national representative surveys in France

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1. Introduction

The advent of the SARS-CoV-2 pandemic has prompted health agencies and researchers around the world to warn against potentially catastrophic consequences on the mental health of populations. In the early stages of the pandemic, a plethora of cross-sectional surveys were launched to assess mental health yielding high rates of depressive symptomatology among young adults and particularly among college students (e.g., (Huang and Zhao, 2020; Wang et al., 2020). Fewer studies have documented changes in rates of depression comparing pre-pandemic data to data collected during the pandemic. Such studies have reported a three-fold increase in depression among adults in the United States in the early stages of the pandemic (Ettman et al., 2020; Twenge and Joiner, 2020). In contrast, a systematic review of longitudinal cohort studies initiated prior to the pandemic and that assessed mental health again during the early stages of the pandemic described significant but small increases in symptomatology (Robinson et al., 2022). Similar conclusions were drawn in a review of studies examining lockdowns (Prati and Mancini, 2021). Some studies have also shown that there were no significant changes in mental health from pre-pandemic to pandemic times, or that initial increases in symptomatology subsequently decreased suggesting that people are resilient and generally adapt to stressful situations (Breslau et al., 2021; Burdzovic Andreas and Brunborg, 2021; Hettich et al., 2022; Johansson et al., 2021).

While it may be tempting to associate potential increases in depression with the disruption caused by the COVID-19 pandemic, it is important to keep in mind that increases in depression had been reported prior to the pandemic, especially among younger generations. For instance, in the U.S., based on data from the National Survey on Drug Use and Health and focusing on adolescents ages 12 to 17, it was estimated that the prevalence of past-year major depressive episode (MDE) had increased from 8.1% in 2009 to 15.8% in 2019, and that this increase was particularly high among girls (Daly, 2022). This increase had previously been

documented in early analyses of these data and showing significant increases in depression from 8.7% in 2005 to 11.3% in 2014 in adolescents and from 8.8% to 9.6% in young adults, ages 18 to 25 (Mojtabai et al., 2016). Relying on data from the Monitoring the Future study, yearly cross-sectional surveys of school-attending adolescents, an increase in depressive symptoms was documented between 1991 and 2018, especially among girls (Keyes et al., 2019). In Germany, and using medical claims data, it was reported that the prevalence of major depression had increased from 12.5% in 2009 to 15.7% in 2017, with the strongest increases observed among males and among adolescents and young adults (Steffen et al., 2020). In France, few studies have documented adolescent mental disorders prior to the pandemic. One survey, conducted on a representative sample of adolescents aged 13 to 14, the Enclass survey, suggested no significant increase in reporting feeling depressed between 2014 and 2018, though an increase in irritability, nervousness and difficulty falling asleep was reported (Léon et al., 2021). Similarly, data from the ESCAPAD survey, a recurring national survey of 17 year olds also reported that the prevalence of probable depression had remained stable between 2014 and 2017 (Observatoire français des drogues et des tendances addictives, 2023). However, results from the same survey showed a significant increase in suicidal thoughts and behaviors among 17 year olds between 2011 and 2017 (Janssen et al., 2019).

Lastly, while there are more studies on the mental health of college students than non-college attending peers, few studies have directly compared the prevalence of mental health problems of college students as compared to same age peers who have joined the workforce or who are not in employment, education or training (NEET). Drawing from large and representative population-based studies focusing on this age group, there is consistent evidence showing that a significant portion of college students reports common mental disorders including mood, anxiety or substance use disorders and that same-aged peers who are not attending school tend to have similar or higher rates of disorders, especially NEETs

(Auerbach et al., 2016; Blanco et al., 2008; Kovess-Masfety et al., 2016). It is therefore crucial to rely on nationally representative samples to understand time trends in major depression in order to identify young adults at highest risk.

Using a national survey that is repeated close to every five years in a large representative sample of residents of metropolitan France, the first objective of this study is to examine time trends in the prevalence of MDE among young adults. The second objective is to examine the prevalence of depression by occupational status comparing students, those employed, and NEETs.

2. Methods

2.1. Survey design and sample

The Health Barometer is a population-based cross-sectional computer-assisted telephone interview (CATI) survey conducted in a nationally representative sample by Santé Publique France to measure behaviors, health determinants and inequalities among the French population using the same methodology (Richard et al., 2018). The present study combined data from the last 4 surveys conducted 2005, 2010, 2017, and 2021, respectively. The surveys relied on a two-stage random sample. First, household selection was performed using randomly generated phone numbers. Second, one individual was randomly selected within each household (Kish, 1949). If a household or respondent refused or could not be contacted, there was no replenishment. In addition, mobile phone numbers were randomly generated to reach individuals who may or may not have a landline. Data were collected by interviewers from a professional survey firm who had received specific training for this survey. The CATI system allowed for direct data capture, skip patterns, and automatic detection of inconsistencies. Consent was obtained over the phone. National surveys conducted by Santé Publique France were exempt from approval by a formal ethical review board. However, they

were required to be approved by the French commission on data privacy and public liberties (Commission Nationale Informatique et Liberté). The cross-survey participation rate was 58.2% in 2005, 52.7% in 2010, 48.5% in 2017 and 44.3% in 2021. Survey waves were combined yielding a total of 101,204 respondents. For the purpose of the present study, the sample was restricted to respondents who were aged 18 to 25 at the time they participated (n=11,933). Of note, while all respondents were asked about depression in the 2005, 2017, and 2021 surveys, major depression was only assessed in a random subsample in the 2010 survey representing one third of respondents. For analysis, the sample was further restricted to respondents in that age group who had responded to major depression variables and who had no missing data on the occupational status, yielding a final sample of n=7,556 respondents for analysis.

2.2.Survey variables

2.2.1. Current occupational status

The sample was categorized into three mutually exclusive categories based on their current occupational status at the time of the survey: employed, students, and NEETs. The employed comprised all persons who reported holding a job, regardless of the nature of the position. Students comprised students attending school or university. NEETs comprised those who reported no current occupation and no current enrollment in any school, training program or college.

2.2.2. MDE in the past 12 months

In each survey, the presence of a DSM-IV-TR and ICD-10 MDE in the prior 12 months was assessed with the World Health Organization's Composite International

Diagnostic Interview – Short Form (CIDI-SF) (Kessler et al., 1998), which has shown satisfactory reliability and internal consistency (Gigantesco and Morosini, 2008). A MDE was defined as endorsing either sadness or anhedonia, at least three additional symptoms (loss of energy, change in weight, difficulties sleeping, difficulties concentrating, self-depreciation, thoughts of death) nearly all day, every day for at least two consecutive weeks associated with functional impairment. The level of severity of the episode was based on the following criteria: mild (sadness or anhedonia + 3 to 4 additional symptoms + mild functional impairment), moderate (sadness or anhedonia + 3 to 4 additional symptoms + moderate functional impairment or 5 to 7 additional symptoms + mild or moderate functional impairment) and severe (sadness or anhedonia + 5 to 7 additional symptoms + severe functional impairment).

2.2.3. Sociodemographic variables

Sociodemographic characteristics used in the present study included sex (male, female), age, region of residence (rural, urban other than Paris, Paris), living situation (living alone, not living alone), and income level. Adjusted household income based on net monthly total income and number of individuals living in household (https://data.oecd.org/hha/household-disposable-income.htm) was categorized in terciles reflecting lowest income (first tercile) to highest income level (third tercile), and undisclosed level of income.

2.3.Data analysis

Data from the 2005, 2010, 2017, and 2021 Health Barometers were pooled, then restricted to respondents 18 to 25 at the time of the survey. Descriptive analyses were used to

present sample characteristics as well as the prevalence of a 12-month MDE (any MDE and by severity level) overall, by sex and occupational status (students vs those employed vs NEETs) using cross-tabulations. Statistical significance was a priori defined by a p-value < .05. Pearson's Chi-squared (Chi2) tests were used to compare proportions (bivariate statistics) and prevalence observed at each time-point. Univariate and multivariate logistic regressions were then performed to identify factors including survey wave, occupational status, and relevant covariates associated with the presence of a 12-month MDE. All assumptions for the use of logistic regression were met. Logistic regressions were performed overall and by severity level (using no depression as the reference). Data were weighted based on the probability of being solicited through the Kish method (that is the ratio of the number of eligible individuals to the number of telephone lines in a household), and to match the structure of the French population with respect to sex, age groups, region of residence, urban unit size, household size and education level, using data from the most recent National Institute of Statistics and Economic Studies (INSEE). All analyses were performed using Stata® (version 15.0 SE).

3. Results

3.1.Sample characteristics

Sample characteristics are summarized in Table 1, and are presented overall and by occupational status. Overall, 15.9% of young adults reported living alone, and only a minority (16.6%) lived in a rural area. The distribution of all variables examined varied significantly by occupational status. For instance, the proportion of males was higher among those employed (56.0%), followed by students (49.3%) and lower among NEETs (45.0%). Similarly, living alone was more common among those employed (18.5%) as compared to students (15.5%) or

NEETs (11.5%). Income level was lowest among NEETs (49.5%), followed by students (41.3%) and those employed (33.7%).

Place Table 1 here

3.2.Prevalence of 12-month MDE and episode severity among young adults

The prevalence of MDE and episode severity is presented in Table 2, by occupational status and by sex. Overall, 13.9% [12.9-14.8] of young adults reported a MDE in the past 12 months: 1.0% [0.7-1.3] a mild episode, 7.8% [7.1-8.6] a moderate and 5.1% [4.5-5.7] a severe episode. Significant differences in the prevalence of MDE was observed by occupational status: NEETs had the highest rate (18.4% [15.5-21.7]) followed by students (14.3% [13.1-15.6]) and those employed (11.0% [9.7-12.5]). Depression severity was also highest among the NEET group, while the proportion of those with mild depression was very low for all three groups. Women had significantly higher rates of MDE as compared to men (18.0% [16.5-19.5] vs 9.9% [8.8-11.1]; p<.001). Increased MDE rates among NEETs followed by students and those employed were observed both among females and males.

Place Table 2 here

3.3.Trends in the prevalence of 12-month MDE between 2005 and 2021 among young adults

The prevalence of MDE among young adults was 10.1% [8.4-12.2] in 2005, 9.7% [7.6-12.2] in 2010 and 11.3% [9.2-12.7] in 2017, reflecting a stable prevalence between 2005 and 2017 (Figure 1). The prevalence nearly doubled in 2021, with 20.9% [19.0-23.0] of MDE, with a significant increase of 9.6 points between 2017 and 2021. At every time point, women

had a higher prevalence of MDE as compared to men. Among women, while the prevalence was stable between 2005 and 2017, a significant increase was observed between 2017 (14.6%, [12.6-17.0]) and 2021 (26.3%, [23.3-29.5], +11.7 points, p<.001). Among men, a first significant increase (+3.4 points) was observed between 2010 and 2017, followed by a second a larger increase between 2017 (7.9%, [6.4-9.7]) and 2021 (15.8%, [13.4-18.5], + 7.9 points, p<.001).

Place Figure 1 here

At each time point, NEET had a significantly higher prevalence of MDE as compared to students or those employed (Figure 2). Among NEET, MDE rates were stable between 2005 and 2017 then doubled between 2017 (14.1%, [10.4-19.0]) and 2021 (28.8%, [22.2-36.6], p<.001). Among students, a first 3.5 points increase was observed between 2005 (9.3%, [7.1-12.2]) and 2017 (12.8%, [10.9-15.0], p<.05), followed by a second and larger increase (+8 points) between 2017 and 2021 (from 12.8%, [10.9-15.0] to 21.0%, [18.5-23.8], p<.001). Finally, among those employed after a stable period from 2005 and 2017, a significant 10.4 points increase was observed between 2017 (7.3%, [5.6-9.4]) and 2021 (17.7%, [14.8-21.0], p<.001).

Place Figure 2 here

When examining the prevalence of MDE by severity status, significant differences were observed in time trends (Figure 3). Mild depression remained stable from 2005 to 2021. In contrast, moderate depression increased between 2017 and 2021 (from 6.7%, [5.7-7.9] to

11.5%, [10.0-13.2], p<.001), as did severe depression (from 3.8%, [3.0-4.8] to 8.4%, [7.1-9.9], p<.001).

Place Figure 3 here

3.4.Multivariate analysis of factors associated with the presence of a 12-month MDE overall and by severity level.

Adjusting for key sociodemographic characteristics (Table 3), NEET remained at increased odds of 12-month MDE (AOR=1.9, 95%CI=1.5-2.5). Factors significantly associated with increased odds of MDE included female sex (AOR=1.9, 95%CI=1.6-2.3), residing in the Paris region (AOR=1.5, 95%CI=1.1-2.0) and the 2021 survey wave (AOR=2.2, 95%CI=1.7-2.8). Age and living alone were not associated with odds of MDE.

Sociodemographic predictors were differentially associated with mild, moderate, and severe depression. When predicting mild depression, the only factor associated with increased odds was female sex (AOR=3.5, 95%CI=1.9-6.3). Occupational status, age, place of residence, income level or living alone were not associated with mild depression.

In contrast with what was observed for mild depression, survey wave and NEET were associated with greater risk of both moderate and severe depression. Living in an urban setting was also associated with moderate but not with severe depression.

Place Table 3 here

4. Discussion

Based on nationally representative data in France, the present analyses show that among young adults: 1) the prevalence of past-year MDE has significantly increased between 2017 and 2021 whereas it had been relatively stable from 2005 to 2017, and 2) significant differences in the prevalence of MDE were found by occupational status whereby those not in employment, education or training were at highest risk overall and particularly among males.

In the wake of the COVID-19 pandemic, the scientific literature suddenly abounded in cross-sectional online surveys conducted in non-probabilistic convenience samples. While such studies may have contributed to highlight potential mental health risks, they may also have contributed to obscure the interpretation of possible changes in mental health during that period. Such studies tended to report very high rates of depressive symptomatology and often attributed such high rates to the ongoing pandemic, yet repeated national surveys applying the same sampling strategy over time and using a structured clinical interview to ascertain the presence or absence of a major depressive episode are in a unique position to identify trends in clinically significant depressive symptomatology over time (Kessler et al., 2022). Such studies involving adults across the lifespan revealed that rates of depression either modestly increased in recent years (Kessler et al., 2022) or did not increase significantly (Knudsen et al., 2021; Ten Have et al., 2023) when comparing recent data collected before vs during the pandemic. However, these studies also specified that younger adults tended to be at greater risk for pandemic-related increases (Knudsen et al., 2021; Ten Have et al., 2023; Winkler et al., 2020), consistent with the present findings. Indeed, some studies investigating the impact of the COVID-19 pandemic have shown the vulnerability of young adults to the experience of anxiety and depression during the early stages of epidemic (Huang and Zhao, 2020; McLafferty et al., 2021; Wang et al., 2020). In describing these findings, some have proposed that social distancing and mandatory lockdowns during the pandemic were associated with an

increased sense of social isolation and loneliness that may have been more sharply felt among young people (Werner et al., 2021). The pandemic has been both directly and indirectly associated with stressors of all levels of severity, ranging from minor stressors to traumatic events. In addition, mandatory lockdowns have been associated with increases in domestic violence (Piquero et al., 2021). A "dose-response" association between number of COVID-related stressors including financial, social and emotional stressors has also been found in relation with stressor related disorders (Abdalla et al., 2021). It is therefore likely that the pandemic contributed to the present findings of a drastic increase in MDE in 2021 among young adults. That said, it is not possible to determine how much of the observed peak is attributable to pandemic-related stressors. Differences observed by occupational status argue in favor of an association between such stressors and major depression. Those who are NEET display higher rates of MDE at each point in time and a sharp increase in 2021. Data from North America shows that the proportion of NEETs itself increased during the pandemic due to the loss of employment particularly among young adults, however this increase was not found in France (Layton, 2022). Conversely, those with depression may be unable to work or to study, may have dropped out of school although the cross-sectional nature of the survey and the lack of information regarding lifetime psychopathology does not allow us to determine whether prior depression affected occupation or the reverse or both. Taken together, the present findings point to the need for public health officials to pay close attention to possible reasons why young adults in the country may find themselves in a situation of being without employment nor education. Preventing school dropout or providing additional programs to reach out to these youths in order to facilitate obtaining gainful employment may be a first step in preventing such difficulties.

An increase in MDE among young adults in 2021 may also reflect a previous increase among adolescents. Consistent with several studies conducted in the U.S. (Daly, 2022; Keyes

et al., 2019; Mojtabai et al., 2016) and in Germany (Steffen et al., 2020), trends describing increases in depression among adolescents have been reported prior to the pandemic. In contrast, French data available prior to the pandemic did not suggest an upward trend regarding depressive symptoms among adolescents, although significant increases in suicidal thoughts and behaviors among 17 year olds between 2011 and 2017 have been reported (Janssen et al., 2019).

Several possible explanations have been proposed to understand increases in the prevalence of depression among young people in recent years. Scientific and technological advances in the past decades have drastically and irrevocably altered every aspect of daily life. The most significant change is that today's 20 year olds -Gen Z- have never known a world without the Internet, and most likely grew up with a tablet or smartphone throughout their childhood and adolescence. An association between depression and time spent on screens has been described among youth (Twenge et al., 2017), as well a related decrease in sleep duration (Twenge et al., 2019). Social media has also been associated with a sense of social isolation (Primack et al., 2017). Relatedly, sedentary lifestyles and insufficient physical activity are known to be associated with depression (Pearce et al., 2022). Furthermore, the prevalence of insufficient physical activity in high-income countries is twice that observed in low-income countries, and has increased in recent years from 31.6% in 2001, to 36.8% in 2016 (Guthold et al., 2018) and even more so in the context of the pandemic, along with increased screen and social media usage. Taken together, these environmental, technological and cultural factors are likely interacting in their contribution to the observed sharp increase in depression. Regardless of the factors implicated in the surge of depression, the country is faced with an unprecedented situation with regard to the mental health of young people which calls for additional work investigating the mechanisms involved in order to develop effective preventive strategies. Considering that young adults tend not to seek specialized mental health

services and display a number of attitudinal barriers such as preferring to handle mental health problems on their own (Janota et al., 2022), efforts are needed to identify those at need of services and facilitating access to care. Such efforts may lie in the stepped care approach increasingly developed in middle and high income countries, as well as the use of smartphones to provide online interventions that may be particularly suited for young generations both to increase intentions to seek help (Ebert et al., 2019), and to reduce depression and anxiety symptomatology (Karyotaki et al., 2022).

Limitations

The present findings should be interpreted in light of several limitations. First, the survey focused on 12-month MDE and did not determine age of disorder onset of lifetime psychopathology which limits the interpretation of the findings. Second, psychomotor agitation or retardation, a symptom of MDE was not assessed. Lastly, while the national Health Barometer surveys have used the same strategy to identify eligible participants in all survey waves included in the present analyses, steady decreases in survey responses rates were observed from 2005 (RR=58.2%) to 2021 (RR=44.3%), a trend similar observed in other high income countries.

Conclusions

A sharp increase in the prevalence of MDE was observed among young adults in 2021. Major depression is most prevalent among NEETs, followed by students, and those employed. Although it is not possible to identify the reasons why these increases are observed, they nonetheless need to prompt research among younger age groups as well as secondary prevention among young adults in order to facilitate access to specialized mental health services and evidence-based treatments. Results from the next Health barometer survey,

scheduled for 2024 will be very informative as it will help understand whether the observed increase was a temporary peak associated with pandemic related stressors or whether increases in depression are sustained.

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Table 1. Sample characteristics

	18-25 year olds n=7556		Employed n=2752		Students n=3807		Not in employment, education or training n=997		Chi2	
	%	95%CI	%	95%CI	%	95%CI	%	95%CI	Chi^2	p
Overall	**100.0**	**-**	**34.5**	**33.2-35.8**	**49.1**	**47.8-50.5**	**16.4**	**15.3-17.6**	**-**	**-**
Survey wave									**20.66**	**.049**
2005	23.7	22.5-25.0	24.4	22.3-26.6	22.7	21.1-34.5	25.0	21.6-28.8		
2010	11.0	10.3-11.9	10.8	9.5-12.2	10.7	9.6-11.8	12.7	10.6-15.3		
2017	33.9	32.7-35.2	31.8	29.8-33.9	35.1	33.3-36.9	34.9	31.3-38.6		
2021	31.4	30.1-32.7	33.0	30.9-35.2	31.5	29.8-33.3	27.4	23.9-31.0		
Sex									**48.17**	**<.0001**
Male	50.9	49.5-52.3	56.0	53.7-58.2	49.3	47.4-51.2	45.0	41.1-48.9		
Female	49.1	47.7-50.4	44.0	41.8-46.3	50.7	48.9-52.6	55.0	51.1-58.9		
Age									**1817.3643**	**<.0001**
18-19	25.3	24.2-26.6	7.4	6.2-8.6	41.1	39.3-43.0	15.8	13.1-18.9		
20-21	27.0	25.7-28.2	19.0	17.2-21.0	33.9	32.1-35.8	22.6	19.4-26.1		
22-23	23.3	22.2-24.5	31.7	29.6-33.9	16.6	15.3-17.9	25.8	22.7-29.3		
24-25	24.4	23.3-25.6	41.9	39.7-44.2	8.4	7.5-9.4	35.8	32.1-39.7		
Lives alone									**30.83**	**<.0001**
No	84.2	83.3-85.0	81.6	80.0-83.1	84.5	83.3-85.6	88.5	86.1-90.6		
Yes	15.9	15.2-16.7	18.4	16.9-20.0	15.5	14.4-16.7	11.5	9.5-13.9		
Income level									**508.09**	**<.0001**
1st tercile	40.0	38.7-41.4	33.7	31.5-36.0	41.3	39.5-43.1	49.5	45.6-53.4		
2nd tercile	21.5	20.4-22.5	34.0	32.0-36.2	15.4	14.2-16.7	13.3	11.2-15.7		
3rd tercile	18.3	17.3-19.4	20.5	18.8-22.4	17.8	16.4-19.2	15.4	13.0-18.2		
Undisclosed	20.2	19.0-21.5	11.8	10.1-13.6	25.5	23.8-27.3	21.8	18.4-25.8		
Region of residence									**72.25**	**<.0001**

Rural	16.6	15.6-17.6	19.8	18.1-21.6	14.3	13.1-15.6	16.6	14.1-19.4
Urban	63.6	62.2-64.9	61.6	59.3-63.9	63.1	61.2-64.9	69.3	65.6-72.8
Paris region	19.8	18.7-21.0	18.6	16.7-20.7	22.6	21.0-24.3	14.1	11.4-17.4

Note: Percentages are weighted. Bold indicates significant chi square tests at p<.05.

Table 2. Prevalence of 12-month MDE and episode severity among 18 to 25 year olds overall and by sex.

		12-month MDE		Episode severity					
				Mild		Moderate		Severe	
	n	%	95%CI	%	95%CI	%	95%CI	%	95%CI
All 18-25 year olds	7556	13.9	12.9-14.8	1.0	0.7-1.3	7.8	7.1-8.6	5.1	4.5-5.7
Current status									
Employed	2752	11.0	9.7-12.5	0.7	0.4-1.3	6.1	5.1-7.3	4.2	3.4-5.2
Students	3807	14.3	13.1-15.6	1.2	0.9-1.7	8.1	7.2-9.2	5.0	4.3-5.8
Not in employment, education or training	997	18.4	15.5-21.7	0.8	0.4-1.8	10.4	8.1-13.1	7.3	5.5-9.6
Chi^2		**39.60**	**p<.0001**	4.08	p=.2762	**22.09**	**p=.0017**	**16.29**	**p=.0062**
Among females	3769	18.0	16.5-19.5	1.5	1.1-2.1	9.6	8.5-10.8	6.9	5.9-7.9
Current status									
Employed	1283	15.4	13.1-18.0	1.3	0.7-2.5	8.3	6.6-10.4	5.7	4.4-7.5
Students	1940	18.6	16.7-20.6	1.8	1.2-2.7	9.9	8.5-11.4	6.9	5.7-8.3
Not in employment, education or training	546	20.6	16.6-25.4	0.9	0.3-2.9	11.2	8.1-15.1	8.6	6.0-12.1
Chi^2		9.00	p=.0675	3.12	p=.4327	4.17	p=.2881	5.48	p=.1862
Among males	3787	9.9	8.8-11.1	0.5	0.3-0.8	6.0	5.1-7.0	3.4	2.8-4.1
Current status									
Employed	1469	7.6	6.1-9.4	0.3	0.1-0.7	4.3	3.3-5.8	3.0	2.1-4.2
Students	1867	9.9	8.4-11.6	0.6	0.3-1.2	6.3	5.1-7.8	3.0	2.3-4.0
Not in employment, education or training	451	15.7	11.8-20.5	0.7	0.3-1.8	9.4	6.3-13.7	5.6	3.5-8.9
Chi^2		**29.31**	**p=.0001**	2.51	p=.3056	**18.26**	**p=.0006**	**9.86**	**p=.0379**

Note: Percentages are weighted. Bold indicates significant chi square tests at p<.05.

Table 3. Multivariate analysis of factors associated with the presence of a 12-month MDE overall and by severity level.

	12-month MDE (n=7536)				Mild episode (n=6594)				Moderate episode (n=7094)				Severe episode (n=6892)			
	n	**%**	**AOR**	**95%CI**	**n**	**%**	**AOR**	**95%CI**	**n**	**%**	**AOR**	**95%CI**	**n**	**%**	**AOR**	**95%CI**
Survey wave		***								***				***		
2005 (ref.)	1694	10.1	- 1 -		1539	1.2	- 1 -		1613	6.1	- 1 -		1592	3.4	- 1 -	
2010	909	9.7	0.8	[0.6-1.1]	838	1.5	1.2	[0.5-3.0]	863	4.9	0.7	[0.4-1.1]	856	3.9	1	[0.6-1.7]
2017	2612	11.3	1	[0.7-1.2]	2334	0.8	0.7	[0.3-1.4]	2500	7,0	1	[0.7-1.3]	2410	4.1	1.1	[0.7-1.5]
2021	2341	20.9	**2.2*** **	**[1.7-2.8]**	1902	1.2	1	[0.5-2.2]	2138	12.7	**2.0*** **	**[1.4-2.8]**	2053	9.6	**2.8*** **	**[2.0-4.0]**
Age																
18-19 (ref.)	1810	13.8	- 1 -		1587	1.3	- 1 -		1698	8.1	- 1 -		1653	5.6	- 1 -	
20-21	1853	13.1	0.9	[0.7-1.2]	1631	1.2	0.8	[0.4-1.7]	1739	7.3	0.9	[0.7-1.2]	1711	5.6	1	[0.7-1.4]
22-23	1867	14.7	1.1	[0.9-1.4]	1625	1.5	1.2	[0.5-2.5]	1756	8.9	1.2	[0.8-1.6]	1692	5.6	1	[0.7-1.6]
24-25	2026	13.9	1	[0.8-1.3]	1770	0.5	0.3	[0.1-1.1]	1921	9,0	1.1	[0.8-1.6]	1855	5.5	1	[0.6-1.5]
Sex		***				***				***				***		
Male (ref.)	3787	9.9	- 1 -		3438	0.5	- 1 -		3638	6.2	- 1 -		3547	3.6	- 1 -	
Female	3769	18,0	**1.9*** **	**[1.6-2.3]**	3175	1.8	**3.5*** **	**[1.9-6.3]**	3476	10.5	**1.7*** **	**[1.4-2.1]**	3364	7.7	**2.2*** **	**[1.7-2.8]**
Status		***								**				**		
Employed (ref.)	2752	11,0	- 1 -		2470	0.8	- 1 -		2621	6.4	- 1 -		2561	4.5	- 1 -	
Students	3807	14.3	**1.4** **	**[1.1-1.7]**	3297	1.4	1.3	[0.6-2.9]	3569	8.6	**1.5** **	**[1.2-2.0]**	3449	5.5	1.2	[0.9-1.7]
Not in employment, education or training	997	18.4	**1.9*** **	**[1.5-2.5]**	846	1,0	1.2	[0.4-3.4]	924	11.3	**2.0*** **	**[1.4-2.9]**	901	8.2	**1.9** **	**[1.3-2.8]**
Lives alone																
No (ref.)	5899	13.5	- 1 -		5181	1.1	- 1 -		5555	8,0	- 1 -		5409	5.4	- 1 -	
Yes	1657	15.9	1.1	[0.9-1.3]	1432	1.2	1.2	[0.6-2.4]	1559	9.8	1	[0.8-1.3]	1502	6.4	1.1	[0.8-1.5]
Income level		***								***						
1st tercile	2845	16.3	1.2	[1.0-1.6]	2433	1.2	1.1	[0.5-2.5]	2659	10,0	1.2	[0.9-1.7]	2561	6.7	1.3	[0.9-1.9]

2nd tercile	1850	14.1	**1.3***	**[1.0-1.7]**	1624	1.2	1.3	[0.5-3.0]	1742	8.6	1.3	[0.9-1.8]	1692	5.5	1.4	[0.9-2.1]
3rd tercile (ref.)	1571	11.5	- 1 -		1401	1,0	- 1 -		1498	7.3	- 1 -		1448	3.9	- 1 -	
Undisclosed	1290	10.8	0.8	[0.6-1.1]	1155	1,0	0.8	[0.3-2.4]	1215	5.6	0.7*	[0.4-1.0]	1210	4.9	1.1	[0.7-1.6]
Region of residence		***				*				***						
Rural (ref.)	1401	10.6	- 1 -		1263	1.1	- 1 -		1324	5.1	- 1 -		1316	5.1	- 1 -	
Urban	4862	14.1	1.2	[0.9-1.5]	4242	0.9	0.7	[0.3-1.5]	4579	8.4	**1.5***	**[1.1-2.1]**	4445	5.9	1	[0.7-1.4]
Paris region	1273	15.9	**1.5****	**[1.1-2.0]**	1089	1.9	1.5	[0.6-3.8]	1191	10.5	**2.1*****	**[1.4-3.0]**	1131	4.9	0.9	[0.6-1.4]

Note: Bold indicates significant odds ratios at p<.05. The level of significance is specified as follows: ***: p<.001; **: p<.01; *: p<.05. Adjusted odds ratios (AOR) are adjusted for all variables presented in the table. No depression is the reference for each of the outcomes considered.

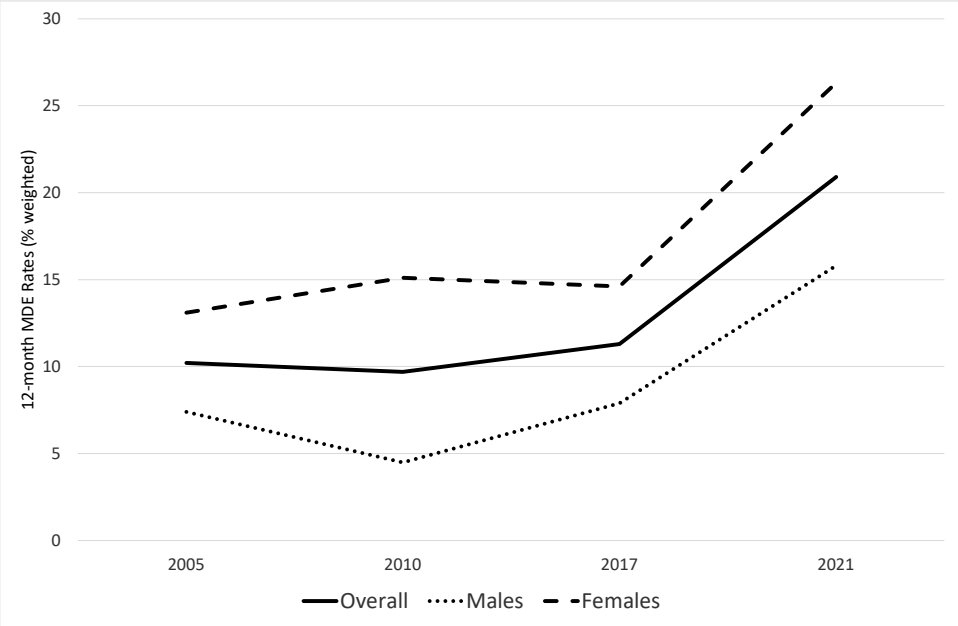


Figure 1. Trends in the prevalence of 12-month MDE among 18 to 25 year olds by sex.

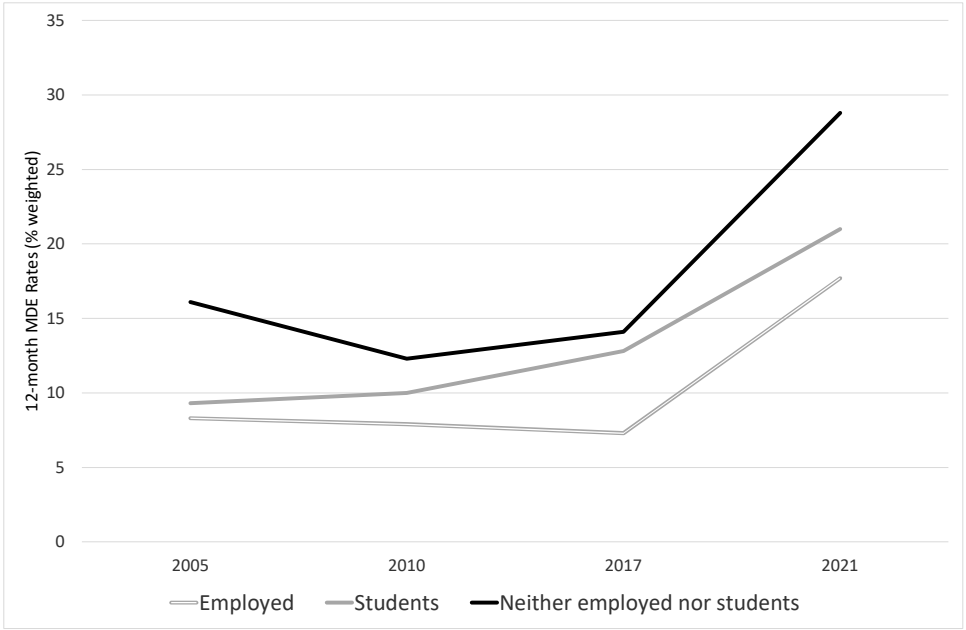


Figure 2. Trends in the prevalence of 12-month MDE among 18 to 25 year olds by occupational status.

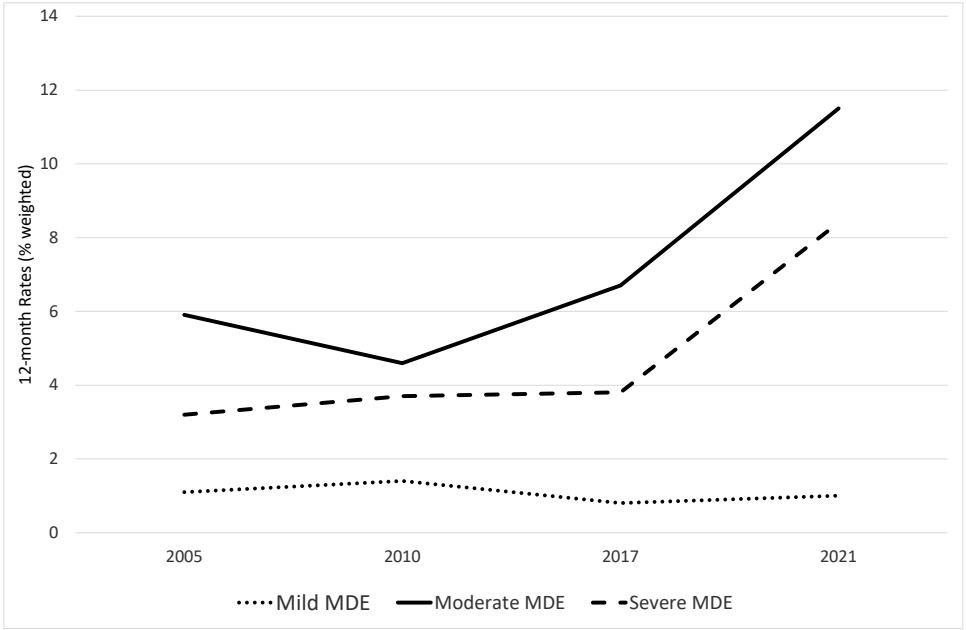


Figure 3. Prevalence of mild, moderate and severe MDE by time period.